

**COMBINATORIAL
GROUP THEORY AND
TOPOLOGY**

EDITED BY

S. M. GERSTEN

AND

JOHN R. STALLINGS

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PREFACE

Combinatorial Group Theory lives in the fertile region between pure group theory and pure topology. The interplay between the logical precision of algebra and the intuitive depths of geometry gives it charm and strength. Lyndon suggests in his article that perhaps there is no exact definition of the subject. Certainly it includes the study of equations over groups (which tends to involve the study of geometric diagrams), much of 3-manifold theory (the Poincaré Conjecture is equivalent to a group-theoretic question), group actions on geometric-combinatorial objects such as trees, the theory of surface automorphisms, and many other developments. As the Walrus said, the time has come to talk of many things.

At Alta Lodge in the spectacular Wasatch Mountains of Utah, on July 15-18, 1984, sixty of us gathered for an intense conference. Roger Lyndon's opening address, in the tradition of Hilbert, offered a score of open problems to guide the field in the future. In the style of the Séminaire Bourbaki, six speakers were assigned topics for expository talks; these were the origins of the papers in this book by Alperin and Bass, Bleiler, Eckmann, Hempel, and Howie. In addition, there were about twenty-five shorter talks on current research, from which we selected the remaining articles here.

Our goal was to produce a book full of ideas, understandable to ourselves and to students, that will open up the future development of this field. We consciously tried to reach a large class of readers, including good graduate students with backgrounds in group theory and topology. We are grateful for the cooperation of the authors who have helped us in pursuing this goal.

It is a pleasure to acknowledge the assistance given us by Roger Lyndon, with whom we frequently consulted in the planning of the conference. In tribute to his work in the field and his insight into directions for future research, his article has a prominent place in this volume.

We are greatly indebted to James Howie and Geoffrey Mess for frequent consultations. In addition, we wish to thank some thirty referees, who by tradition must remain anonymous, for their valuable aid.

Special thanks are due to Marty Jones of Alta Lodge and to Ann Reed for helping make the conference run smoothly. Finally we acknowledge with thanks and appreciation the financial support of the University of Utah and the National Science Foundation.

STEVE GERSTEN
JOHN STALLINGS

**Combinatorial Group Theory
and Topology**

PROBLEMS IN COMBINATORIAL GROUP THEORY

Roger Lyndon

0. *Introduction*

Steve Gersten asked me to give a talk like Hilbert gave in Paris in 1900. I said I'd be happy to, but pointed out that I'm no Hilbert. Merzlyakov [117] came to my rescue with the following suggestion: "Rather than waiting [for a new Hilbert] group theorists have come to the more prosaic idea of a Present-day collective Hilbert." So I have appealed to some of you for amendments to a provisional outline, for which I thank you, and I hope this exonerates me of any charge of arrogance in presenting this biased survey.

What is Combinatorial Group Theory? This term acquired official status as the title of the book of Magnus-Karrass-Solitar. The first sentence of the History of Combinatorial Group Theory by Chandler-Magnus [28] says: "Combinatorial Group Theory may be characterized as the theory of groups which are given by generators and relations...." (But compare Magnus-Karrass-Solitar with Coxeter-Moser, Generators and Relations for Discrete Groups.) This hardly does justice to the goals or methods of the subject. On other occasions Chandler-Magnus speak of "group theory with the exception of Lie groups and of group representations and linear groups," but this seems to both include and exclude too much.

Maybe we could define Combinatorial Group Theory to be Very Low Dimensional Topology. In fact, group theorists, like others, will attack whatever problems interest them with whatever tools they have at hand, so perhaps Combinatorial Group Theory is just a state of mind. Maybe it

is distinguished by a reluctance to make the great supposedly simplifying assumptions of Commutativity, Linearity, and either Finiteness or Continuity, or, put more positively, a relish for the combinatorial core of a problem. Altogether, it seems fruitless to try to define an elephant we have barely touched.

I intended to talk about the history of the subject, but Chandler-Magnus have let me off that hook. I do not need to tell anyone that the major source and strength of Combinatorial Group Theory has been Topology, wherein we tentatively include Discontinuous Groups, from Poincaré on, with an assist from the more or less abstract or axiomatic side, beginning with Cayley, through the influence of Finite Groups and of problems from Logic. These influences will be evident, and, if my discussion appears biased against topology, it is because as a non-topologist I am reluctant or unable to tell you what you know better than I. Altogether, though, I was agreeably surprised by the homogeneity of the subject, if somewhat inconvenienced by the many interconnections among the various approaches and problems in the subject.

My account is also biased for the most part toward rather recent work, and to pathways to the future more than monuments to the past. The space devoted to a subject should not be taken as a measure of the importance attached to it. Likewise, the mention of a name or citation of a paper is often more to draw attention to it than to bestow an honor. References are illustrative or suggestive, and should usually be followed by 'and others.' I have not cited papers that are well known or easily accessible (with a few exceptions), in particular papers more than about five years old, or listed in the bibliography of Lyndon-Schupp. Incomplete references indicate lack of knowledge.

Despite the unity of the subject, as a first step toward linearization I have divided it into seven unequal and gerrymandered sections. It took some cutting and pasting to bring my list of problems to twenty, a modest three fewer than Hilbert. Even so, many are really 'problem areas,' of the classical form: 'What can be said about?.'

1. *Properties of free groups, equations in and over groups*

The most 'abstract' and 'axiomatic' of our problems is a folklore problem of Alfred Tarski.

PROBLEM 1. *Do all nonabelian free groups have the same elementary theory?*

The elementary theory of a class of groups is the set of all sentences in first order logic (with symbols for equality and group composition but, emphatically, excluding set theory) that are true in all groups of the class. By contrast, among free *abelian* groups those of rank at most 2 are distinguished by the property that there exist elements a and b such that, for all x , one of x , xa , xb , xab is the square y^2 of some element y . See V. Dyson [47].

In this connection, R. Vaught posed the following test problem: In a free group, does $x^2y^2 = z^2$ always imply $xy = yx$? This was proved by Lyndon and generalized extensively. The method was close to that used by H. Zieschang in studying automorphisms of surface groups and Fuchsian groups.

A much earlier theorem of Frobenius (1895) deals with the number of solutions of the equation $x^n = 1$ in a finite group; for a survey of this problem see H. Finkelstein [57]. See also Finkelstein-Mandelberg [58].

Equations *over* groups entered Combinatorial Group Theory in a paper of B. H. Neumann (1943), where he showed that, for a positive integer n and an element g of a group G , the equation $x^n = g$ has as many solutions as desired in some group H containing G . Higman-Neumann-Neumann extensions are a generalization of the fact that the equation $t^{-1}g_1t = g_2$, for g_1 and g_2 in G , has a solution t in a group H containing G if and only if g_1 and g_2 have the same order.

The central problem in this area is the Kervaire-Lauderbach problem, which we state as follows.

PROBLEM 2. *If G has a presentation $G = \langle X : R \rangle$ and $H = \langle X \cup t : R \cup w \rangle$ is obtained by adding one new generator and one defining relation, when does the inclusion $X \rightarrow X \cup t$ induce an injection of G into H ?*

In simpler language, for $g_1, \dots, g_n$ in G , when does an equation $w(g_1, \dots, g_n, t) = 1$ have a solution t in some group H containing G ?

We have seen that $t^{-1}g_1tg_2^{-1} = 1$ has no solution if g_1 and g_2 have different orders. Gerstenhaber-Rothaus showed that if the sum of the exponents on t in w is not 0, and if G can be embedded in a compact connected Lie group, then a solution always exists. Rothaus [144] later improved the condition on G to local residual finiteness. The sufficient condition of local indicability has been studied by J. Howie [94]. The group $G = \langle a_1, \dots, a_4 : a_{i+1}^{-1}a_ia_{i+1} = a_i^2, i \text{ modulo } 4 \rangle$ of G. Higman satisfies none of the known sufficient conditions for the solvability of an equation (with non-zero exponent sum) over G . Without the exponent condition, probing by Lyndon [113] suggests that the problem is difficult even for G a finite cyclic group. See also Brodskii [19, 20] and Short [152].

The result of Gerstenhaber-Rothaus raises the following question.

PROBLEM 2a. *If the sum of the exponents on t in w is not 0, does the equation $w = 1$ always have a solution?*

So much for equations over groups. Equations in groups are exemplified by the Vaught problem. This prompted Lyndon to ask about solutions of an equation $w(a_1, \dots, a_n, t_1, \dots, t_m) = 1$ in a free group G with basis $a_1, \dots, a_n$. In the case $m = 1$ of one unknown he obtained a set of words containing parameters as exponents which, subject to conditions on the parameters, give precisely the set of solutions. For example, $t^{-1}a_1ta_1^{-1} = 1$ has exactly the solutions $t = a_1^n$ for all integers n . This was substantially improved and extended, but without definitive result. Recently Makanin [115] has given an algorithm that associates with an equation $w = 1$, as above, an integer N such that, if any solution exists, there

exists one with total length of the t_i at most N . This settles the question of existence of solutions and provides an algorithm for finding one if it exists, but leaves open the following problem.

PROBLEM 3. *Given an equation over a free group, find an algebraic description of the set of all solutions.*

See Howie [93, 95, 97], Ozhigov [130].

A special case of this problem, for free groups and other groups, is the Substitution Problem (for free groups, the Endomorphism Problem): Does an equation $w(t_1, \dots, t_m) = g$ have a solution? Wicks [169] showed that an element g of a free group is a commutator if and only if, relative to any basis, g can be written in the form $g = abca^{-1}b^{-1}c^{-1}$ *without cancellation*. This result has been extended substantially by C. C. Edmunds [51, 52] and L. P. Comerford, Jr. [34, 35] and, using topological methods, by M. Culler [37] and by Goldstein-Turner [73, 74]. Related methods have been used by M. Scharlemann [146] and P. E. Schupp [148]; see also E. Rips [140], and P. Hill and S. J. Pride [90].

PROBLEM 4. *Let $w(a_1, \dots, a_n)$ be a word in the free group F with basis $a_1, \dots, a_n$. Is there an algorithm which, given g in F , decides if there exist $t_1, \dots, t_n$ in F such that $w(t_1, \dots, t_n) = g$?*

The Substitution Problem has been studied extensively for finite simple groups and for various classical infinite groups. See Finkelstein [57], Finkelstein-Mandelberg [58], Lyndon [112], Mycielski [122], and also Ehrenfeucht-Fajtlowicz-Malitz-Mycielski [53].

2. Automorphisms of groups

The general linear group $GL(n, \mathbf{R})$ is the group of automorphisms of the free $\mathbf{R}$ -module of rank n . The automorphism group $GL(2, \mathbf{Z})$ of the free $\mathbf{Z}$ module, or free abelian group, A_2 of rank 2, and also, in par-

ticular, the modular group $\text{PSL}(2, \mathbb{Z})$, have been much studied. But it is a giant stride to the study of the automorphism group $\text{Aut } F_n$ of the free nonabelian group F_n of rank n .

Nielsen used automorphisms of free groups to great advantage, and obtained a finite presentation for $\text{Aut } F$. J. McCool recovered this presentation by different methods, which enabled him to obtain finite presentations for the stabilizers of finite sets of elements. His method is based on that used by Whitehead to decide whether two finite sequences of elements of a free group are equivalent under some automorphism. A related problem of Whitehead, to decide whether two finitely generated subgroups are equivalent under an automorphism, has been solved recently by Gersten [68], using new methods. By the same methods he has proved a conjecture of P. Scott that the subgroup of elements fixed by an automorphism of a finitely generated free group is finitely generated. J. L. Dyer and Scott had shown earlier that the set of fixed points of a finite subgroup of $\text{Aut } F$ is a free factor of F . There has been some study of the structure of a single automorphism, inducing the dream, or nightmare, of a 'Jordan structure theorem.'

Considerable work on the structure of $\text{Aut } F$, related mainly to its action on various naturally arising characteristic subgroups and their quotients, has been done by S. Andreadakis and by S. Bachmuth-H. Mochizuki.

PROBLEM 5. *Determine the structure of $\text{Aut } F$, of its subgroups, especially its finite subgroups, and its quotient groups, as well as the structure of individual automorphisms.*

If N is a normal subgroup of a free group F and $\text{Aut}_N F$ is its stabilizer, then $\text{Aut}_N F$ induces a group of automorphisms of $G = F/N$. Nielsen showed that for the usual presentation of a surface group G the group $\text{Aut}_N F$ maps onto $\text{Aut } G$. G. Rosenberger [143] and H. Zieschang have studied this 'lifting problem,' especially for Fuchsian groups. See also S. J. Pride and A. D. Vella [137].

PROBLEM 6. *If $G = F/N$, F free, what subgroups of $\text{Aut } G$ are images of subgroups of $\text{Aut } F$?*

The mapping class group M of a surface can be identified essentially with the group of outer automorphisms (automorphisms modulo inner automorphisms) of its fundamental group G . The results of J. McCool give M (or, directly, $\text{Aut } G$) as the fundamental group of a finite complex, which, unfortunately, exhibits too many natural symmetries to make it accessible to present day computation. W. Thurston (see A. Hatcher-W. Thurston [88]) has given a quite different method for obtaining a finite presentation of the mapping class groups, and, for the orientable case, B. Wajnryb [168] has used this method to obtain presentations that are reasonably concise but not entirely perspicuous.

PROBLEM 7. *Obtain finite presentations for the mapping class groups that are at once usably concise and yet in which both the generators and the relations have fairly obvious geometrical meanings.*

For example, Hatcher-Thurston say: "... all relations follow from relations supported in certain subsurfaces, finite in number, of genus at most 2."

The following is an obvious addendum.

PROBLEM 7a. *The same for all Fuchsian groups.*

3. *Morphisms of trees*

The prefix 'auto' is omitted as a salute to the recent work of Gersten [62, 64-72] and Stallings [158, 159] on morphisms in the category of graphs.

A basic paper of J. Tits [164] initiates the study of the group $\text{Aut } T$ of automorphisms of a tree T . He shows that, if $\text{Aut } T$ leaves invariant no proper subtree and no end of T , then the subgroup G generated by

all stabilizers of branch points is a simple group, while the quotient $(\text{Aut } T)/G$ is a free product of groups of order 2 and infinite cyclic groups. This frequently cited work deserves to be extended.

A great deal is coming to be known about certain very special but very remarkable groups of automorphisms of trees, introduced by N. Gupta and S. Sidki [84, 85, 86, 153, 154]. If A is an 'alphabet' with a prime number p of letters, then the monoid $T = A^*$ of all finite words, ordered by left divisibility, is a tree T . Generalizing a construction of R. I. Grigorchuk [79], Gupta and Sidki show that certain easily described 2-generator subgroups G of $\text{Aut } T$ have remarkable properties: they are 'Burnside groups,' that is, infinite 2-generator p -groups (with elements of unbounded order); they contain isomorphically all finite p -groups; they are residually finite, and all their proper quotient groups are finite. These groups are remarkable not only for these properties, but also because they are quite 'concrete' and accessible to detailed study.

PROBLEM 8. *Study the structure of the automorphism groups of trees and of their subgroups.*

For example, do these groups have Sylow subgroups?

Certain naturally arising instances of groups acting on trees encountered by J.-P. Serre appear to have led to the Bass-Serre theory of graphs of groups, with their associated groups acting on trees. This method has become a standard tool in the study of infinite groups, especially as obtained by amalgamated product and HNN-extension, and with regard to subgroup theorems. Earlier, Lyndon, in seeking to unify cancellation arguments based on Nielsen transformations in the proofs of the Nielsen-Schreier and Kurosh Subgroup Theorems and of the Grushko-Neumann Theorem, introduced axiomatically characterized length functions on groups. I. M. Chiswell showed that this theory, for integer valued functions, is essentially equivalent to the Bass-Serre theory of groups acting on trees. J. W. Morgan and P. B. Shalen [120], following

work of R. C. Alperin and K. N. Moss [4] on real valued length functions, have played off these two theories to obtain new proofs of two theorems of Thurston.

In connection with the Subgroup Theorems, we note that Rosenberger, Zieschang, and Karrass-Solitar all obtained refinements of the basic Subgroup Theorem of Hanna Neumann for amalgamated products. In particular, Karrass and Solitar introduced tree products, which agree with a special case of the Bass-Serre graph products. They also introduced polygonal products: a group G is generated by vertex groups, with the edge groups amalgamated. (This group G differs in a small but significant way from the corresponding Bass-Serre group; the Karrass-Solitar definition is natural in the context of presentations of certain geometrically constructed groups, while that of Bass-Serre is natural for a graph of complexes.) Polygonal products were motivated by the recognition that the Picard group $\mathrm{PSL}(2, \mathbb{Z}[i])$ can be obtained from four very small groups at the vertices of a square by amalgamating subgroups associated with the sides of the square. See, for example, B. Fine [55, 56] and A. Brunner, M. L. Frame, Y. W. Lee, N. J. Wielenberg [24]. Square products are studied also in a paper of D. Ž. Djoković [42], which, although failing of its main objective, contains an extensive study of groups acting on cubic trees. See also Djoković-G. L. Miller [43]. A. Brunner, Y. W. Lee, and N. J. Wielenberg [25] have used polygonal products to obtain elegant descriptions of various 3-dimensional Euclidean groups. M. W. Davis [40] has used a similar construction in a more abstract context to obtain generalized Coxeter groups by sewing together infinitely many copies of an abstract polytope by identifications at vertices according to specified orthogonal groups; in this way he obtains aspherical manifolds of dimension $n \geq 4$ not covered by Euclidean space. There are also various abstractly defined generalizations of symmetric groups, braid groups, and Coxeter groups; K. I. Appel and P. E. Schupp [5, 6] have used small cancellation theory to study certain such groups.

PROBLEM 9. *The various constructions mentioned above appear to open the way to a more comprehensive theory of the structure of infinite groups.*

We mention a curious result concerning the question of accessibility by D. E. Muller and P. E. Schupp [121], that a finitely generated group is virtually free if and only if it has a context-free word problem and is accessible.

See also W. Dicks [41], J-C. Hausmann [89], A. Karrass-A. Pietrowski-D. Solitar [108], Stallings [158], and M. D. Tretkoff [165].

4. *Burnside groups*

W. Burnside asked if every finitely generated torsion group is finite. This is true for groups of certain small exponents, for linear groups, and for analogous Lie rings. In 1960 Golod proved the existence of what we shall call *Burnside groups*, finitely generated infinite torsion groups. In 1964 P. S. Novikov and S. I. Adian showed that $B(m, p)$, the free group on $m \geq 2$ generators in the variety of groups of prime exponent p , is infinite for sufficiently large p ; see [1]. We have mentioned that Grigorchuk [79] gave a very simple construction for an infinite 3-generator 2-group, and that N. D. Gupta and S. Sidki extended this construction to obtain infinite 2-generator p -groups for all primes p . Before Grigorchuk, S. V. Aleshin [3] had constructed groups like Grigorchuk's in terms of automata, which Y. I. Merzlyakov [118] showed to be essentially equivalent to those of Grigorchuk.

A. Y. Olshanskii [124, 125, 126, 128, 129] used small cancellation theory to construct infinite 2-generator groups, with solvable conjugacy problem, in which all nontrivial proper subgroups are isomorphic cyclic groups, all infinite cyclic or all of order p , for some prime $p > 10^{75}$. Olshanskii's groups are Tarski Monsters, groups all of whose proper subgroups have smaller cardinality. S. Shelah [151], also using small cancellation theory, had constructed earlier an uncountable Tarski Monster. E. Rips

(see [140]) has obtained results, not wholly published, parallel to results of Olshanskii, using similar methods.

The groups $B(m,p)$ of Novikov-Adian are universal in the sense of being as large as possible given the number m of generators and the exponent p , and thus they have a simple definition, but at the expense of being complicated to work with. Despite this Adian [1] has obtained considerable detailed information about them. The groups of Olshanskii are more economical, as images of the Novikov-Adian groups, but their definition is much more complicated, their presentation being dictated naturally by the problem at hand; from this it is easier to establish their crucial properties. The groups of Gupta-Sidki, although not of finite exponent, combine the virtues of being very easily defined and relatively easy to work with.

The Burnside problem may be viewed as a test problem, of more interest in terms of methods than of results. Nonetheless, the natural generality of the Novikov-Adian groups together with Adian's detailed results on them, the fact that Olshanskii's groups of exponent p are infinite finitely generated simple groups, and the fact that the Gupta-Sidki groups arise as groups of automorphisms of trees and have interesting properties, all lead to the conclusion that Burnside groups are of considerable intrinsic interest. (Perhaps there was a time when finite p -groups were considered as of interest only as Sylow groups.)

PROBLEM 10. *Develop a general theory of Burnside groups, including a detailed study of certain particular such groups.*

Despite enormous work, mainly for very small and very large prime values of n , the following problem is far from solved.

PROBLEM 10a. *For which pairs m and n is the group $B(m,n)$ finite?*

Another natural problem seems to be folklore, and little more.

PROBLEM 10b. *Is every finitely presented torsion group finite?*

I have found no reference to this problem in print. Gupta has pointed out that all those who have constructed infinite Burnside groups have been at pains to point out that their groups are not finitely presented. Scott has told me that he and, independently, W. Jaco, had recognized that the existence of a finitely presented infinite torsion group would follow from the existence of a 3-manifold whose nonabelian fundamental group had an infinitely generated center.

For a survey of Burnside groups see Gupta [82].

5. *Generators and relations*

From the point of view of presentations, the simplest groups after free groups are the one-relator groups. These groups have been studied extensively, partly because the surface groups are one-relator groups, and partly because they share some of the accessibility of free groups while exhibiting considerable individual complexity. Magnus, giving credit to Dehn's topological insight, solved the word problem for one-relator groups, and obtained other results by the same method, notably the Freiheitssatz. Although his method looks rather easy, and perhaps even obvious from the algebraic side, in retrospect, it has yet to be incorporated into a general theory in an entirely satisfactory way, especially from the topological point of view. In connection with the Freiheitssatz, see Lyndon, and B. Baumslag-Pride, and J. Howie [93]. It is striking that, despite the solution of the conjugacy problem for a fair variety of groups, the conjugacy problem for one-relator groups remains unsettled fifty years after Magnus' solution of the word problem.

PROBLEM 11. *Is the conjugacy problem solvable for all one-relator groups?*

Note that B. B. Newman has solved this problem for all one-relator groups with torsion.

Recently A. Juhász [103, 104, 105, 106] has developed new methods in small cancellation theory that solve the conjugacy problem for certain

groups, including a new special class of one-relator groups. Other recent work, especially of S. J. Pride [133, 134, 135, 136] and of Pride together with P. Hill [90] and with P. Hill and A. D. Vella [91], has extended small cancellation theory to derive considerable information about the structure of one-relator groups and their subgroups. For example, Pride has shown that a one-relator group with torsion can have only finitely many isomorphism types of nonfree two-generator subgroups. See also G. Baumslag [9] and H. D. Hurwitz [102].

PROBLEM 11a. *Determine further the structure of one-relator groups.*

Although small cancellation theory seems now to be the best tool for studying one-relator groups and, especially, their conjugacy problem, if it turns out that there are one-relator groups with unsolvable conjugacy problem, it would seem to require very considerable ingenuity on the side of logic to construct one.

A second question arising simply in terms of generators, initially without reference to relations, is that of the rate of growth on a group. If a group G is generated by a finite set X , let $g(n)$ be the number of elements of G represented by words in X of length at most n . J. Milnor, who introduced these ideas in connection with differential geometry, asked if the asymptotic rate of growth of $g(n)$, which does not depend on the choice of X , is always either polynomial (as for a free abelian group) or exponential (as for a nonabelian free group), and, with J. A. Wolf, established this conjecture for finitely generated solvable groups; for these M. Gromov [80] (see also A. J. Wilkie-L. van den Dries [170]) showed that the growth is polynomial just in case the group contains a nilpotent group of finite index. Recently Grigorchuk has shown that one of his groups is neither polynomial nor exponential.

J. W. Cannon [27] showed that for certain Fuchsian groups and Coxeter groups, with a natural choice of generating set X , the growth generating function $G(z) = \sum g(n)z^n$ is a rational function, and characterized its zeros and poles. M. Benson [14] showed that $G(z)$ is rational for finite

extensions of free abelian groups of finite rank. See also M. Grayson [77]. Grigorchuk [78] considered a function $H(z) = \sum h(n)z^n$ for H a subgroup of a free group F , where $h(n)$ is now the number of elements of H of minimal length n relative to a basis X for F , and showed that $H(z)$ is a rational function. For H normal and $G = F/H$, he obtains the spectral radius r_G of a random walk on G in terms of the radius of convergence of $H(z)$, and proves the following. To given $\epsilon > 0$ and integers M and N , there corresponds an integer L with the property that, if G has a presentation with M generators and $n \leq N$ relators, all of length at least L , which satisfies a standard small cancellation condition, then $|r_G - \sqrt{2M-1}/M| < \epsilon$, and hence G is not amenable. Adian uses these ideas to show that his infinite groups $B(m,p)$ are not amenable. C. Series [149, 150] uses similar methods to study the distribution of the set of limits of random walks on Fuchsian groups. See also H. Bass [8], W. J. Floyd [59], and P. Wagreich [167].

PROBLEM 12. *There is clearly much to be done in determining the possible growth functions of groups and in relating them to properties of groups.*

A similarity between the formula for the growth function of a free product and that for the characteristic suggests a connection between these two concepts, which is borne out in some of Cannon's examples. It appears that further understanding is needed of the dependence of the detailed structure of the growth function on the choice of generating set, and, with it, on properties of the set of relations.

A long standing question is that of the significance of the deficiency of a presentation, the excess of the number of relations over the number of generators. Deficiency of a finite group has been much studied. The deficiency can be viewed as a truncation of a Poincaré series of a resolution of the presentation, and seems related to the Euler-Poincaré characteristic. See B. Baumslag-S. J. Pride [10, 11], K. S. Brown [21, 22] (and

the review of [21] by K. W. Gruenberg [81]), I. M. Chiswell [29], M. Edjvet [49], J. G. Rattcliffe [138], N. S. Romanovskii [142] and R. Stöhr [160]. We pose only a rather vague problem.

PROBLEM 13. *Extend and relate the theories of the deficiency, the rate of growth, and the Euler-Poincaré characteristic. In particular, what influence does the deficiency have on the structure of an infinite group?*

This seems an appropriate point to speak of certain problems of a directly topological origin which have led to equivalent or related problems that can be stated in purely group theoretic terms. I do not feel it is my part to do more than mention these problems.

PROBLEM 14. *The Poincaré conjecture.*

See G. A. Swarup [161]. Among the many problems in group theory stimulated by the Poincaré conjecture, that of J. J. Andrews and M. L. Curtis is possibly the best known.

PROBLEM 15. *Let the trivial group have a balanced presentation $(X:R)$, where $|X| = |R| < \infty$. Can this presentation be reduced to a trivial presentation by a succession of transformations of the following kinds:*

1. *Nielsen transformations of X ;*
2. *Nielsen transformations of R ;*
3. *Replacing an element of R by a conjugate;*
4. *Tietze transformations introducing (or deleting) a new element x of X together with a new relator r in R defining x ?*

See R. Craggs [36], W. Metzler [119], and C. P. Rourke [145].

A geometrically finite group G is one possessing a finite $K(G,1)$, and its geometric dimension is then the least dimension of such a $K(G,1)$.

Geometrically finite groups, along with many related concepts and problems, are discussed in G. Baumslag-E. Dyer-A. Heller [12]; in connection with a related problem of Baumslag-Dyer-Heller see T. Bortnik [18]. The equality of geometric dimension and cohomological dimension has been proved with one exception.

PROBLEM 16 (The Eilenberg Problem). *If G has cohomological dimension 2, must it also have geometric dimension 2?*

We remark that Baumslag-Dyer-Heller examine algebraically closed groups, using, as does S. D. Brodskii [19], a definition that is natural but which differs from the original definition given by W. R. Scott.

A third problem from topology with many group theoretic connections is Whitehead's asphericity problem.

PROBLEM 17. *Is every subcomplex of an aspherical 2-complex aspherical?*

This problem has received considerable attention recently. See R. Brown-J. Huebschmann [23], I. M. Chiswell-D. J. Collins-Huebschmann [30], Collins-Huebschmann [31], M. Gutierrez-J. G. Ratcliffe [87], J. Howie [96], Huebschmann [99, 100, 101], and A. J. Sieradski [155].

The problems above touch on the large area of questions of a homological nature, of which we mention only one more. That is the study of Poincaré duality groups; for this see R. Bieri [2] and, especially, the talk by B. Eckmann.

We cannot resist introducing small cancellation theory with a quotation from J. Nielsen (1912), see [123].

“While in the solution of the generation problem one arranges elements in sequence in all possible ways – one could speak of a linear problem – one represents the [relators] in the group graph by polygons and in the treatment of the identity problem one puts these together in a map. This latter problem has, therefore, in an expression

of Dehn's, 'one more dimension' than the first. Whether this is a real or only an apparent difference in level ... remains undecided."

Small cancellation theory has become a useful and ubiquitous tool. Applied small cancellation theory is exemplified by recent work of Juhász, Olshanskii, Pride, and Rips, who have also, as well as, for example, Do Long Van [44], J. Howie-S. J. Pride [98], and J. Perraud [132], greatly extended the methods of pure small cancellation theory.

The theory, as developed so far, could be described as follows. From 'local' hypotheses, on 'small' subcomplexes of the (2-dimensional) Cayley complex (Gruppenbild) of a presentation (initially, on the stars of faces), one derives properties of 'large' subcomplexes (singular discs, spheres, annuli, tori, ...) or of the entire complex. There is an evident analogy, or connection, with corresponding differential-integral concepts, for example, in small cancellation theory the formulas relating 'arc length,' 'curvature,' and 'area.' (Two papers in the 40's by C. Blanc [16] and F. Fiala [54], developing and applying complex function theory on planar graphs do not seem to have been pursued further. There is more recent work by P. Cartier on harmonic analysis on trees, and also work on harmonic analysis on free groups.) Thus, as well as Dehn's original application of small cancellation theory in the hyperbolic plane to logical decision problems, from which the theory arose, and of the more or less obvious connections with topology (for example, asphericity) and with purely group theoretic 'structure problems' (for example, Pride's work on subgroups), there are connections of an analytic nature, concerning rate of growth, random walks, analysis on groups, etc.

PROBLEM 18. *Extend small cancellation theory, especially in accordance with the connection or analogy with analysis, and unify the special extensions and applications noted above.*

6. Geometric groups

Much of combinatorial group theory arose from the study of Fuchsian groups, if one includes surface groups, and there is no need to mention

the role of hyperbolic groups in the study of higher dimensional manifolds. In the study of Fuchsian groups, especially surface groups, there seems to be a long tradition, beginning with Dehn and Magnus, and with Nielsen, and continuing through the work of Reidemeister and Zieschang and, in an extreme form, two papers of A. H. M. Hoare, A. Karrass, and D. Solitar on subgroups of Fuchsian group, which could be described, somewhat invidiously, as a more or less conscious attempt to free the subject from analysis. This tendency is not entirely a matter of prejudice, in view of the increasing difficulty of applying analytic methods in higher dimensions and in more abstract situations, notably to 'geometric' groups defined combinatorially without any direct reference to an analytic structure, as mentioned in Section 3.

To a non-analytic mind it seems a great mystery that combinatorial and analytic methods often lead to the same results; it happens often that objects definable combinatorially are realizable analytically. A modest example is the combinatorially regular tessellations of the n -sphere and n -space for $n \geq 2$, and a more impressive one is the definition by Zieschang, E. Vogt, and H. D. Coldewey of Fuchsian groups as automorphism groups of planar 2-complexes. So perhaps we should not reject analysis, but seek to assimilate it.

A striking example of such assimilation is J. Cannon's work based in part on the fact that, if the Cayley complex for a discontinuous hyperbolic group is realized in the natural way in hyperbolic space, then the concepts of combinatorial (word) and metric hyperbolic geodesic agree about as well as could be hoped for. This provides a new approach to small cancellation theory as well as opening up other vistas.

Something along the same line appears, with a different purpose, in the papers of C. Series. A geometrically combinatorial approach appears in a number of papers by A. L. Edmonds, J. E. Ewing, and R. S. Kulkarni (e.g. [50]) in which they study, for example, torsion free subgroups of finite index in Fuchsian groups. Somewhat in the same spirit is the

rather extensive use by a number of workers of coset graphs in the study of Riemann surfaces and of subgroups and quotient groups of Fuchsian groups and other groups. See, for example, recent work of J. L. Brenner-R. C. Lyndon and of W. W. Stothers.

PROBLEM 19. *Develop a unified combinatorial theory of a suitably comprehensive class of 'geometric' groups.*

See also Floyd-Hoare-Lyndon [60] and C. L. and M. D. Treutkoff [165].

7. *Algebraic representations of groups*

We are now far distant from logic and topology and approaching the forbidden vastnesses of representation theory, Lie theory, and group rings. But, despite the interdiction by Chandler-Magnus, we must mention Magnus' representation of free groups in associative rings, Lie rings, and matrix rings. The first and central example here is Magnus' representation of a free group F in the power series completion A of a free associative ring, carrying with it a representation of the descending central quotients of F by the dimension modules of the free Lie ring associated with A . We refrain from mentioning all but one of the many extensions of these ideas.

The connection between the commutator structure of a group, especially a p -group, and Lie theory, although not as perfect as for continuous groups, has been highly developed, especially by P. Hall and his followers. Closely related to the Magnus representation is the free differential calculus of R. H. Fox, originating in knot theory, but with a homological interpretation along the lines of the work of Reidemeister.

Any group G is naturally embedded in its integral group ring ZG . Let Δ be the fundamental ideal, the kernel of the augmentation map $ZG \rightarrow Z$. The n -th dimension subgroup $D_n(G)$ of G is then defined to be $G \cap (1 + \Delta^n)$, the group of all g in G such that $g \equiv 1$ modulo Δ^n . Magnus (1935) showed that if G is a free group, then $D_n(G) = \gamma_n G$, the

n -th term of the descending central series of G ; from $\bigcap \Delta^n = 0$ it follows that $\bigcap \gamma_n G = 1$.

It was conjectured that $D_n(G) = \gamma_n G$ for all groups G and all positive integers n , and this conjecture is easily reduced to the case that G is a finite p -group. However, Rips (1972) exhibited a 2-group G with $\gamma_4 G = 1$ but $D_4(G) \neq 1$. This very special counterexample, using the only even prime, seems to leave the door open for a possible rehabilitation of the problem. See I. B. S. Passi [131], J. A. Sjogren [156], K-I. Tahara [162, 163], and the talk by Gupta.

Fox (1953) introduced the groups $F(n, R) = F \cap (1 + J\Delta^n)$, where R is a normal subgroup of the free group F and J is the kernel of the natural map $ZF \rightarrow Z(F/R)$. The dimension subgroups of $G = F/R$ are represented analogously by the groups $D(n, R) = F \cap (1 + J + \Delta^n)$. By the result of Magnus, $F(n, F) = D_{n+1}(F) = \gamma_{n+1} F$, and Magnus showed also that $F(1, R) = \gamma_2 R$. Enright showed that $F(2, R) = \gamma_2(R \cap \gamma_2 F)\gamma_3 R$, and Gupta and Passi derived properties of $F/F(n, R)$ from a matrix representation. For these results see the survey article by Gupta, A problem of R. H. Fox, *Canad. Math. Bull.* 24(1981), 129-136.

PROBLEM 20. *Obtain intrinsic descriptions of the Fox subgroups $F(n, R)$ and the dimension subgroups $D(n, R)$ in terms of the commutator structure of the free group F and its normal subgroup R .*

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Added in proof, February 1, 1985. A letter from G. S. Makanin and R. I. Grigorchuk, containing many valuable comments on the manuscript of this paper, was received too late to incorporate in the text. They propose the following very reasonable additions to the list of problems: (1) (Tarski) Decidability of the elementary theory of free groups; (2) Finiteness of the Burnside groups $B(m, 2^n)$; (3) The conjugacy and equivalence problems for tame knots.

They point out that Problem 4 as stated is solved by the cited result of Makanin. This problem should perhaps be construed as asking for a simpler and more explicit description of the set of all solutions of an equation of the special form $w(x_1, \dots, x_n) = a$.

They have provided the following four important additional references; note that the first two of these papers contain extensive bibliographies.

Adian, S. I. Studies on the Burnside problem and related questions. Trudy Mat. Akad. Nauk SSSR 168(1984), 171-196.

Adian, S. I., Makanin, G. S. Studies on algorithmic questions in algebra. Trudy Mat. Akad. Nauk. SSSR 168(1984).

Grigorchuk, R. I. On Milnor's problem of group growth. Dokl. Akad. Nauk SSSR 271 (1983); Soviet Math. Dokl. 28(1983), 23-26.

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Suščanskii, V. I. Periodic p -groups of permutations and the unrestricted Burnside problem. Dokl. Akad. Nauk SSSR 247(1979).

Added in proof, February 20, 1985. Cameron Gordon has pointed out that the solution to the equivalence problem for knots is contained in the paper by G. Hemion, "On the classification of homeomorphisms of 2-manifolds and the classification of 3-manifolds," Acta Math. 142(1979), 123-155.

POINCARÉ DUALITY GROUPS OF DIMENSION TWO ARE SURFACE GROUPS

Beno Eckmann

1. *Introduction*

We first explain the terms appearing in the title, and add some general comments.

1.1. A *surface group* G is a group isomorphic to the fundamental group $\pi_1(\Sigma_g)$ of a closed surface Σ_g , orientable or not, of genus $g \geq 1$. Such a group admits a presentation

$$G = \langle x_1, y_1, \dots, x_g, y_g \mid [x_1, y_1] \cdots [x_g, y_g] = 1 \rangle$$

in the orientable case,

$$G = \langle z_0, z_1, \dots, z_g \mid z_0^2 z_1^2 \cdots z_g^2 = 1 \rangle$$

in the non-orientable case.

We will also use the concept of a *surface group-pair* $(G; \{S_0, S_1, \dots, S_m\})$: it consists of the (free) fundamental group of a closed surface of genus $g \geq 0$ with $m+1$ disks removed, $m \geq 0$ (but ≥ 1 if the surface is a sphere), together with the $m+1$ infinite cyclic subgroups generated by the boundary circles of these disks. Surface group-pairs have presentations

$$G = \langle t_1, t_2, \dots, t_m, x_1, y_1, \dots, x_g, y_g \rangle,$$

$$S_0 = \langle t_1 t_2 \cdots t_m [x_1, y_1] \cdots [x_g, y_g] \rangle, \quad S_j = \langle t_j \rangle \text{ for } j = 1, \dots, m$$

in the orientable case, $m+g > 0$;

$$G = \langle t_1, t_2, \dots, t_m, z_0, z_1, \dots, z_g \rangle$$

$$S_0 = \langle t_1 t_2 \dots t_m z_0^2 z_1^2 \dots z_g^2 \rangle, \quad S_j = \langle t_j \rangle \quad \text{for } j = 1, \dots, m$$

in the non-orientable case, $m \geq 0, g \geq 0$.

The "lowest" cases of surface group-pairs are

$$G = \langle t_1 \rangle, \quad S_0 = \langle t_1 \rangle, \quad S_1 = \langle t_1 \rangle$$

and

$$G = \langle z_0 \rangle, \quad S_0 = \langle z_0^2 \rangle.$$

1.2. A *Poincaré duality group* G of dimension n , in short a PD^n group, is a group fulfilling Poincaré duality in (co-) homology for all coefficient ZG -modules A with respect to the formal dimension n and to a certain G -action on the additive group of integers $\tilde{Z}$:

$$H^i(G; A) \cong H_{n-i}(G; \tilde{Z} \otimes A), \quad i \in \mathbb{Z}.$$

Here $\tilde{Z} \otimes A$ is a ZG -module by diagonal action, and the isomorphisms are natural in the ZG -modules A . As we will see the dimension n and the ZG -module structure of $\tilde{Z}$ are determined by G (the PD^n -group being orientable or non-orientable according to whether the action is trivial or not).

The definition above is, of course, analogous to the Poincaré duality valid for all closed n -dimensional manifolds X , the coefficients A being $\pi_1(X)$ -modules. If X is aspherical then the (co-) homology of X is isomorphic to that of $G = \pi_1(X)$ so that, in that case, G is a PD^n -group. It is not known whether in general the converse is true; i.e., whether a PD^n -group is necessarily isomorphic to the fundamental group of a closed n -dimensional aspherical manifold.

Since the universal covering of the surface Σ_g , $g \geq 1$, is $\mathbb{R}^2$ the surface Σ_g is aspherical. Thus the surface groups in 1.1 above are PD^2 -groups. The *Theorem* formulated in the title states that the converse

is true, thus solving the problem in the case $n = 2$. The proof of that *Theorem* has been achieved in several steps contained in a series of papers by the present author and various collaborators; these papers were written partly with other objectives in mind. The conference organizers have asked the author to present a survey, as complete as possible, of that proof. We do so and use the opportunity to simplify some of the arguments.

1.3. There are, in fact, different techniques and methods involved in that proof. They belong roughly speaking to the following three areas of ideas:

- (I) Homological algebra, homology of groups.
- (II) Structure and splitting theorems for groups (Stallings-Bass-Serre and others).
- (III) Ranks of projective modules and Euler characteristic (Hattori-Stallings-Bass-Kaplansky).

The papers leading to or containing the steps of the proof are as follows. In the field (I) by Robert Bieri and the author [2], [4], [5]; in (II) by Heinz Müller and the author [12], [14]; in (III) by Peter Linnell and the author [10], [11].

1.4. For compact manifolds-with-boundary (∂ -manifolds), in particular for the closed surfaces with disks removed, one has the well-known “relative” Poincaré duality. The surface group-pairs listed in 1.1 above fulfill such a relative Poincaré duality, of dimension 2. To formulate this in a precise way, relative (co-) homology groups for pairs of groups $(G; \{S_0, S_1, \dots, S_m\})$ have to be considered, cf. Section 2.3 below; this yields the concept of a PD^n -pair of groups. An important step in the proof of the *Theorem* will be to show that all PD^2 -pairs of groups are surface group-pairs (“Relative Theorem,” Section 3.2).

1.5. We mention here two corollaries of the *Theorem*.

COROLLARY 1.1 (cf. [12], [11]). *All Poincaré-2-complexes are homotopy equivalent to closed surfaces (of genus ≥ 0).*

COROLLARY 1.2. *Let G be a torsion-free group containing a surface group S as subgroup of finite index; then G is also a surface group.*

Indeed, a homological argument (see Section 2.2) shows that G is a PD^2 -group. This corollary is a special case of the “Nielsen realization conjecture” proved by Kerckhoff [Annals of Math. 117 (1983), 235-265]. The special case above was established by Eckmann-Müller [12] before Kerckhoff’s proof, and before our Theorem on PD^2 -groups had been completely settled.

1.6. This survey is organized as follows. Section 2 contains general preliminaries on duality and relative duality groups. In Section 3 we assume that the PD^2 -group fulfills a certain “splitting” property and show that the *Theorem* can then be reduced to the Relative Theorem; both that reduction and the proof of the Relative Theorem are given on the basis of general splitting arguments explained in Section 4. In Section 5 we show, by quite different methods involving ranks of finitely generated projective modules, that any PD^2 -group fulfills the splitting assumption.

2. Duality groups

2.1. The group G is called a *duality group of dimension* $n > 0$ with respect to a dualizing ZG -module C , in short a D^n group, if one has isomorphisms

$$H^i(G; A) \cong H_{n-i}(G; C \otimes A)$$

for all $i \in \mathbb{Z}$ and all ZG -modules A ; they are assumed to be natural in A , and $C \otimes A$ is endowed with the diagonal G -action. If $C = \mathbb{Z}$ as an Abelian group, one has Poincaré duality (cf. 1.2), i.e., G is a PD^n -group.

From the definition it follows that $H^1(G; A)$ commutes with direct limits in A . This is possible only if G admits a projective resolution

$\cdots \rightarrow P_i \rightarrow P_{i-1} \rightarrow \cdots \rightarrow P_0 \twoheadrightarrow Z$ with all P_i finitely generated over ZG (by the Bieri-Eckmann-Brown finiteness criterion, see [3] and [7]).

Furthermore one easily checks that $C \otimes ZG \cong C_0 \otimes ZG$ (where C_0 is the Abelian group underlying C) is an induced module, and thus $H^i(G; ZG) = 0$ for all $i \neq n$; as for $H^n(G; ZG)$, it is isomorphic to $H_0(G; C \otimes ZG) = C$, and this is a (right) ZG -module isomorphism. The dualizing module C is thus determined by G ; it is easily seen to be torsion-free as an Abelian group.

The cohomology dimension $\text{cd } G$ is clearly $\leq n$, and by the above it is $= n$; hence the integer n is also determined by G , and G admits a finitely generated projective resolution of finite length (equal to n); such groups are said to be of type FP.

Summarizing we see that a D^n -group G fulfills

- (1) G is of type FP.
- (2) $H^i(G; ZG) = 0$ for $i \neq n$ and $H^n(G; ZG)$ is torsion-free,
- (3) $\text{cd } G = n$.

It has been proved by Bieri-Eckmann [2] that, conversely, a group fulfilling (1) and (2) is a D^n -group with dualizing module $C = H^n(G; ZG)$.

We note that $\text{cd } G = n$ implies that G is torsion-free.

2.2. As an application, let G be torsion-free and S a subgroup of finite index. By Serre's theorem (see e.g. [8], p. 190) $\text{cd } G = \text{cd } S = n$. Clearly G admits a finitely generated free resolution over ZG if and only if S does over GS ; hence G is of type FP if and only if S is. Moreover $H^i(S; GS) \cong H^i(G; \text{Hom}_S(ZG, GS)) \cong H^i(G; GS \otimes_S ZG) = H^i(G; ZG)$, and it follows that G fulfills (1) and (2) above if and only if S does, with the "same" dualizing module C .

Thus G is a D^n -group if and only if S is a D^n -group, and the dualizing modules are isomorphic as Abelian groups. In particular, G is a PD^n -group if and only if S is.

1) REMARKS: *Dimension 1.* G is a D^1 -group if and only if it is finitely generated free. It is a PD^1 -group if and only if it is infinite cyclic.

2) *Subgroups of infinite index.* For PD^2 -groups G , Strebel [17] has proved, by homological methods, that for a subgroup S of infinite index in G one has $cd S \leq n-1$. For $n=2$ it follows that S is a free group; this is the PD^2 -analogue of a fact well known for surface groups.

2.3. We briefly recall *relative duality* for group pairs (for details see Bieri-Eckmann [4], [5]). A group pair $(G, \underline{S})$ consists of a group G and a family $\underline{S} = \{S_j, j \in I\}$ of subgroups, not necessarily distinct. For any subgroup $S \subset G$ one writes ZG/S for the G -module whose underlying Abelian group is freely generated by the cosets xS , with G -action by left multiplication. Relative (co-) homology is defined by means of the "augmentation kernel" $\Delta = \ker \{ \oplus_j ZG/S_j \xrightarrow{\varepsilon} Z \}$ where $\varepsilon(xS_j) = 1$ for all $x \in G$ and $j \in I$:

$$H_i(G, \underline{S}; A) = H_{i-1}(G; \Delta \otimes A),$$

$$H^i(G, \underline{S}; A) = H^{i-1}(G; \text{Hom}(\Delta, A)),$$

A being a G -module, $\otimes$ and Hom equipped with diagonal G -action.

A duality pair of dimension n with dualizing module C is a pair $(G, \underline{S})$ fulfilling

$$H^i(G; A) \cong H_{n-i}(G, \underline{S}; C \otimes A)$$

and

$$H^i(G, \underline{S}; A) \cong H_{n-i}(G; C \otimes A).$$

In the (orientable) Poincaré duality case, $C = Z$, each of these isomorphisms implies the other one and the first one becomes

$$H^i(G; A) \cong H_{n-i-1}(G; \Delta \otimes A);$$

i.e., $(G, \underline{S})$ is a PD^n -pair of groups if and only if G is a D^{n-1} -group

with dualizing module Δ . Relative exact sequences show that $\underline{S}$ must be a finite family of PD^{n-1} -groups $S_0, S_1, \dots, S_m$.

In particular, if $(G, \underline{S})$ is a PD^2 -pair then G is finitely generated free and $\underline{S}$ is a finite family of infinite cyclic groups. The duality isomorphisms yield $H^2(G, \underline{S}; \mathbb{Z}G) = \mathbb{Z}$, $H^1(G, \underline{S}; \mathbb{Z}G) = 0$, $H^1(G; \mathbb{Z}G) = \Delta$. As shown in [4] the only important homological property is $H^2(G, \underline{S}; \mathbb{Z}G) = \mathbb{Z}$; indeed, if G is finitely generated free and $\underline{S}$ a finite family of infinite cyclic groups it characterizes PD^2 -pairs.

3. PD^2 -groups splitting over a finitely generated subgroup

3.1. A group G is said to split over the subgroup H if either (α) G is a non-trivial amalgamated free product $G = G_1 *_H G_2$, or (β) G is an HNN-extension $G = G_1 *_H \mathbb{P}$. Two cases are of special importance:

1) H is finitely generated, 2) H is finite. We recall that Stallings' structure theorem [15], [16] for finitely generated groups tells that 2) holds if and only if $H^1(G; \mathbb{Z}G) \neq 0$.

PROPOSITION 3.1. *The assertion of the Theorem holds if the PD^2 -group G splits over a finitely generated group L .*

The proof of Proposition 3.1 makes strong use of the "simultaneous splitting theorem" (in short: SST) for groups and subgroups, established by Heinz Müller [14]. The SST is a refinement of the relative version (Swan [18], Swarup [19]) of Stallings' structure theorem; it deals with splittings over finite subgroups H , and since in our case G is torsion-free only $H = 1$ will occur. A short outline of SST and its application in the present context will be given in Section 4 below.

The application of SST to PD^2 -groups (α) $G = G_1 *_L G_2$, or (β) $G = G_1 *_L \mathbb{P}$, with L finitely generated, is as follows. Since $H^1(G; \mathbb{Z}G) = 0$, L is $\neq 1$. The index of L in G is infinite; by Strebel's theorem [17] one has $\text{cd } L \leq 1$, and hence L is (finitely generated) free. We only describe the case (α), the case (β) being similar.

If in $G = G_1 *_L G_2$ the rank of L is > 1 one has, by virtue of SST (see Section 4.4, A)), splittings

$$G_1 = H_1 * H_2, \quad L = L_1 * L_2 \quad \text{with} \quad L_i \subset H_i, \quad i = 1, 2,$$

or

$$G_1 = H * \langle q \rangle = H *_{1,q}, \quad L = L_1 * qL_2^{-1}q \quad \text{with} \quad L_1, L_2 \subset H.$$

The first possibility yields

$$G = H_1 *_L (H_2 *_L G_2);$$

if $L_1 \neq H_1$ then G splits over L_1 , and if $L_1 = H_1$ then G splits over L_2 . Thus G splits over a subgroup whose rank is less than that of L .

The second possibility yields

$$G = (H *_L G_2) *_{L_2, q^{-1}},$$

so G splits over L_2 , again of rank less than that of L .

Thus we are always reduced to the case where G splits over an infinite cyclic subgroup C as (a) $G = G_1 *_C G_2$, or (b) $G = G_1 *_{C,p}$.

Since C is a PD^1 -group we can apply the general homological arguments of [4], Theorem 8.1 and 8.3. It follows that in the case (a) the group pairs (G_1, C) and (G_2, C) and in the case (b) the group pair $(G_1, \{C, p^{-1}Cp\})$ are PD^2 -pairs. Now the *Relative Theorem* below tells that these pairs are surface group-pairs (see 1.1) corresponding to closed surfaces with one disk, or two disks respectively, removed.

In (a), $G = G_1 *_C G_2$ is the fundamental group of the closed surface obtained by identifying the boundary circles; in (b), $G = G_1 *_{C,p}$ is the fundamental group of the closed surface obtained by joining the two boundary circles by a tube.

3.2. RELATIVE THEOREM. *Any PD^2 -pair $(G; \{S_0, \dots, S_m\})$ is a surface group-pair.*

Again the proof uses SST, in addition to the properties of PD^2 -pairs mentioned in 2.3. We proceed by induction on the rank of the finitely generated free group G .

If that rank is 1, i.e., G infinite cyclic $C = \langle c \rangle$, one has $H^1(G; ZG) = Z$ since $G = C$ is a PD^1 -group, and $H^1(G; ZG) = \Delta$ since G is a duality group of dimension 1 with dualizing module Δ . The exact sequence

$$\Delta \longrightarrow \bigoplus_j ZG/S_j \longrightarrow Z$$

then yields $\bigoplus_j ZG/S_j = Z \oplus Z$; this is possible only if $\underline{S} = \{C, C\}$ or $\underline{S} = \{\langle c^2 \rangle\}$. Thus the pair $(G, \underline{S})$ is either $(C, \{C, C\})$ or $(C = \langle c \rangle, \{\langle c^2 \rangle\})$; i.e., we obtain precisely the lowest orientable case $g = 0$, $m = 1$, or the lowest non-orientable case $g = 0$, $m = 0$ of the presentation list of surface group-pairs in 1.1.

If the rank of G is > 1 , SST (see Section 4.4, B)) yields splittings (a) $G = G_1 * G_2$ with $S_0 = \langle g_1 g_2 \rangle$, $1 \neq g_j \in G_j$, while the S_j for $j > 0$ are (conjugate to) subgroups of G_1 and G_2 , say $S_1, \dots, S_k \subset G_1$, $S_{k+1}, \dots, S_m \subset G_2$; or (b) $G = G_1 * \langle p \rangle = G_1 *_{1,p}$ with $S_0 = \langle pq_1 p^{-1} g_2 \rangle$, $g_1, g_2 \in G_1$, while $S_1, \dots, S_m$ are (conjugate to) subgroups of G_1 . We will deal with (a) only, the case (b) being similar. We can write

$$G = (G_1 * \langle g_2 \rangle) *_{\langle g_2 \rangle} G_2.$$

Then $S_0 \subset G_1 * \langle g_2 \rangle$. The pair $(G_2; \{\langle g_2 \rangle, S_{k+1}, \dots, S_m\})$ is a PD^2 -pair by Theorem 8.1 of [4]; note that the case $\langle g_2 \rangle = G_2$ needs a special argument, and the pair must then be $(\langle g_2 \rangle, \{\langle g_2 \rangle, \langle g_2 \rangle\})$, cf. [4] p. 517. Similarly the pair $(G_1, \{\langle g_1 \rangle, S_1, \dots, S_k\})$ is a PD^2 -pair. By induction they are surface group-pairs, and one easily checks that so is $(G; \{S_0, S_1, \dots, S_m\})$.

4. Simultaneous splitting of groups and subgroups

4.1. We consider throughout this section a finitely generated group G which splits over a finite subgroup K as $G = G_1 *_{\mathbf{K}} G_2$ or $G = G_1 *_{\mathbf{K}, \mathbf{p}} G_2$, i.e., with $H^1(G; \mathbb{Z}G) \neq 0$, cf. Section 3.1. Given a finite family $\{S_1, \dots, S_m\}$ of subgroups of G let $N(G; S_1, \dots, S_m)$ be the intersection of the kernels of the restriction maps $\text{res}_j : H^1(G; \mathbb{Z}G) \rightarrow H^1(S_j; \mathbb{Z}G)$, $j = 1, \dots, m$. By the relative version of the structure theorem (Swan [18], Swarup [19]) $N(G; S_1, \dots, S_m) \neq 0$ if and only if there is a splitting of G such that the S_j are conjugate to subgroups of G_1 or G_2 ; this will be the case in the following and only such splittings will be considered.

Let T be a further subgroup of G , and assume that T is finitely generated. $\{x_\nu\}$ denoting a set of coset representatives of $G \bmod T$, we consider the restriction map

$$H^1(G; \mathbb{Z}G) \xrightarrow{\text{res}} H^1(T; \mathbb{Z}G) \cong \bigoplus_{\nu} H^1(T; \mathbb{Z}T) x_\nu.$$

The minimal number of non-zero components of $\text{res}(c)$ for all $0 \neq c \in N(G; S_1, \dots, S_m)$ is called the *weight* $n(T)$ of T with respect to G (and to $S_1, \dots, S_m$). Note that $n(T) = 0$ if and only if $N(G; T, S_1, \dots, S_m) \neq 0$; i.e., if there is a splitting of G with $T \subset G_1$ or G_2 .

4.2. The simultaneous splitting theorem (SST) established by H. Müller [14] (actually for more general G and T) concerns the case $n(T) > 0$. It can be formulated roughly as follows, its full content being in fact more complicated.

There is a tree Γ on which G acts with finite edge-stabilizers and with proper subgroups of G as vertex-stabilizers such that Γ/G has one edge (and that each S_j , $j = 1, \dots, m$ stabilizes a vertex of Γ); and Γ contains a subtree Γ_T invariant under T with Γ_T/T having at most $n(T)$ edges.

EXAMPLES. 1) Let G be torsion-free, and $n(T) = 1$. Then one has the following possibilities:

- (1) $G = G_1 * G_2$, $T = T_1 * T_2$, $T_1 \subset G_1$, $T_2 \subset G_2$.
- (2) $G = G_1 * \langle p \rangle = G_1 *_{1,p}$, $T = T_1 * pT_2p^{-1}$, $T_1, T_2 \subset G_1$.
- (3) $G = \langle p \rangle$, $T = \langle p \rangle$, $S_1 = \dots = S_m = 1$ or $m = 0$.

2) Let G be torsion-free, T infinite cyclic, and $n(T) = 2$. Then one has the following possibilities (cf. [14])

- (1) $G = G_1 * G_2$, $T = \langle g_1 g_2 \rangle$, $1 \neq g_i \in G_i$, $i = 1, 2$.
- (2) $G = G_1 * \langle p \rangle = G_1 *_{1,p}$, $T = \langle pg_1p^{-1}g_2 \rangle$, $g_1, g_2 \in G_1$.
- (3) $G = \langle p \rangle$, $T = \langle p^2 \rangle$, $S_1 = \dots = S_m = 1$ or $m = 0$.

4.3. We restrict ourselves to some remarks concerning the proof of SST. We write Z_2 for $\mathbb{Z}/2\mathbb{Z}$ and use Z_2G , Z_2T instead of ZG , ZT ; one easily checks that this yields the same weight $n(T)$. Then $H^1(G; Z_2G)$ can be interpreted as group of all subsets of G which are "almost invariant" under translation without being G or $\emptyset$; almost invariant means invariant except for finite sets. Namely, writing $\overline{Z_2G}$ for $\text{Hom}(ZG, Z_2) = \prod_{x \in G} Z_2x$ and ϵG for $\overline{Z_2G}/Z_2G$ (arbitrary modulo finite subsets of G), the exact sequence

$$H^0(G; \overline{Z_2G}) \rightarrow H^0(G; \epsilon G) \rightarrow H^1(G; ZG) \rightarrow H^1(G; \overline{Z_2G}) = 0$$

yields

$$(\epsilon G)^G / (Z_2G)^G \cong H^1(G; Z_2G).$$

The restriction map $H^1(G; Z_2G) \xrightarrow{\text{res}} \oplus_{\nu} H^1(T; ZT)x_{\nu}$ is then given by $U \mapsto U \cap Tx_{\nu}$ for all ν where U is a non-trivial almost invariant set.

The components $U \cap Tx_v$ are almost T -invariant. By the techniques of Dunwoody [9] and Swarup [19] these almost invariant sets yield the theorem.

4.4. There remains to compute the weights of those subgroups which occur in the proofs of Proposition 3.1 (subgroup L) and of the "Relative Theorem" in 3.2 (subgroup S_0).

A) We again restrict ourselves to the case (a) in the proof of Proposition 3.1. Thus G is a PD^2 -group with $G = G_1 *_L G_2$, L free of rank > 1 . We claim that $n(L)$ with respect to $(G_1; \emptyset)$ or to $(G_2; \emptyset)$ is equal to 1. By example 1) in 4.2 this yields the required simultaneous splittings of G_1 and L .

To prove the claim we consider the exact Mayer-Vietoris sequence

$$\cdots \rightarrow 0 \rightarrow H^1(G_1; ZG) \oplus H^1(G_2; ZG) \xrightarrow{(\text{res}_1 - \text{res}_2)} H^1(L; ZG) \xrightarrow{\delta} H^2(G; ZG) \rightarrow \cdots$$

and note that the weight is not 0 since res_1 and res_2 are injective. $H^1(L; ZL)$ is free Abelian of infinite rank since L is free of rank > 1 . Thus the restriction of δ to $H^1(L; ZL)$ cannot be injective, $H^2(G; ZG)$ being $= Z$; i.e., the intersection $\text{im}(\text{res}_1, -\text{res}_2) \cap H^1(L; ZL)$ is $\neq 0$. On the other hand, if both $n(L)$ with respect to $(G_1, \emptyset)$ and to $(G_2, \emptyset)$ are > 1 , the image $\text{res}_1(c_1) - \text{res}_2(c_2)$ of $0 \neq (c_1, c_2) \in H^1(G_1; ZG) \oplus H^1(G_2; ZG)$ cannot lie in $H^1(L; ZL) \subset H^1(L; ZG)$; this is seen by looking at the lengths of elements with respect to coset representatives of $G \bmod G_1$ and G_2 . Thus $n(T)$ with respect to, say, $(G_1, \emptyset)$ is $= 1$, and we obtain the simultaneous splitting.

B) In 3.2, $(G, \underline{S})$ is a PD^2 -pair, $\underline{S} = \{S_0, S_1, \dots, S_m\}$ with $m \geq 0$ and all S_j infinite cyclic, and we have to consider the case where the free group G is of rank > 1 . The claim is that $n(S_0)$ with respect to $(G, \{S_1, \dots, S_m\})$ is $= 2$. By example 2) in 4.3 this yields the required splittings.

The exact relative cohomology sequence of $G \bmod \underline{S}$ is

$$0 \rightarrow (G, \underline{S}; ZG) \rightarrow H^1(G; ZG) \xrightarrow{r} \bigoplus_{j=0}^m H^1(S_j; ZG) \xrightarrow{\delta} H^2(G, \underline{S}; ZG) \rightarrow 0,$$

where r denotes the map with components res_j , $j = 0, 1, \dots, m$. The PD^2 -pair properties tell that the first term is 0, the last isomorphic to Z . If S_0 (or any proper subset) is omitted from the family $\underline{S}$, the last term becomes 0 and the first term must be $\neq 0$ (cf. [4], Section 11); i.e., the intersection $N = N(G; S_1, \dots, S_m)$ of the $\ker \text{res}_j$, $j = 1, \dots, m$ is non-zero. The weight $n(S_0)$ is the minimal number of components in $H^1(S_0; ZG) = \bigoplus_{\nu} H^1(S_0; ZS_0)_{x_{\nu}} \cong \bigoplus_{\nu} Zx_{\nu}$ of $\text{res}_0(c)$ for all $0 \neq c \in N$; note that $\ker \text{res}_0 \cap N = 0$.

Now $r(N) = (\text{res}_0(N), 0, \dots, 0) = (H^1(S_0; ZG), 0, \dots, 0) \cap \ker \delta$. One easily checks that δ restricted to any summand Zx_{ν} of $H^1(S_0; ZG)$ is bijective. So obviously the minimum number of components of elements $\neq 0$ in $\text{res}_0(N)$ is 2, which proves the claim.

5. The first Betti number of a PD^2 -group

5.1. In order to conclude the proof of the Main *Theorem* we have to show that a PD^2 -group, without any further assumption, splits over a finitely generated subgroup. This is guaranteed by the following proposition whose proof will be given in 5.2 and 5.3 below. Recall that PD^n -groups are of type FP so that Betti numbers $\beta_i(G) = \text{rank } H_i(G; Z)$ are defined.

PROPOSITION 5.1. *If G is a PD^2 -group then $\beta_1(G) > 0$.*

From this it follows that $H_1(G; Z)$, the abelianized group G , contains at least one infinite cyclic summand C , and thus G admits a factor group $\cong C$. By a result of Bieri-Strebel ([6], Theorem A) this implies that G splits as $G = G_1 *_{L, p}$ over a finitely generated subgroup L (as shown in [6], this holds for any group G of type FP_2 , i.e., admitting a projective resolution which is finitely generated in dimensions

≤ 2 , and having an infinite cyclic factor group). The splitting is constructed explicitly, the generator p being any element projecting onto a generator of C .

5.2. For the proof of Proposition 5.1 we use the homological Euler characteristic $\chi(G) = \beta_0(G) - \beta_1(G) + \beta_2(G)$ of G . By the Euler-Poincaré formula $\chi(G)$ is equal to the alternating sum of the ranks of the free Abelian groups $Z \otimes_G P_i$ for any FP-resolution over ZG

$$0 \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow Z.$$

We recall that in the orientable case $\beta_0(G) = \beta_2(G) = 1$ and $\beta_1(G) =$ even; in the non-orientable case $\beta_0(G) = 1$, $\beta_2(G) = 0$ (the proofs are the same as for closed surfaces). Thus in both cases the claim of Proposition 5.1 is $\chi(G) \leq 0$. If G is a non-orientable PD^2 -group, let G_0 be the orientable subgroup of index 2. By the multiplicative property of the Euler characteristic (valid for groups of type FP, cf. [8], §IX.6) $\chi(G_0) = 2\chi(G)$; hence $\chi(G_0) \leq 0$ implies $\chi(G) \leq 0$, and we are reduced to the orientable case.

G being an orientable PD^2 -group we choose a resolution

$$(1) \quad 0 \rightarrow P \rightarrow ZG^d \xrightarrow{\delta} ZG \rightarrow Z$$

with P finitely generated projective over ZG . Applying $\text{Hom}_G(-, ZG)$ to (1) we get the sequence

$$(2) \quad 0 \leftarrow P^* \xleftarrow{\delta} ZG^d \leftarrow ZG \leftarrow 0$$

where $P^* = \text{Hom}_G(P, ZG)$ is finitely generated projective. Since $H^i(G; ZG) = 0$ for $i \neq 2$ and $H^2(G; ZG) = Z$ we obtain the exact sequence

$$(3) \quad Z \leftarrow P^* \xleftarrow{\delta} ZG^d \leftarrow ZG \leftarrow 0,$$

which is another FP-resolution for G ; for $\chi(G)$ it yields